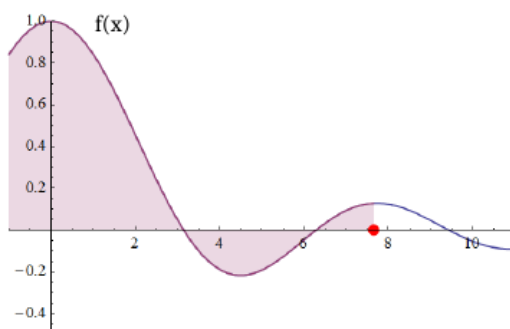


Problem 6

Problem 6

Problem 1 Given the following graph of $y = f(x)$, let $g(x) = \int_{-1}^x f(t) dt$.



- (a) Is g continuous? Why or why not?

Explanation. $g(x)$ represents the area function (or the signed area between the curve $y = f(x)$ and the x -axis from -1 to x). By the FTC(I), $g(x)$ is continuous.

- (b) Is g differentiable? Why or why not?

Explanation. Similarly by the FTC(I) the function g is differentiable, and moreover:

$$\frac{d}{dx} (g(x)) = \frac{d}{dx} \left(\int_{-1}^x f(t) dt \right) = f(x)$$

- (c) Where does g achieve its absolute maximum and minimum values? Where does g achieve any local extreme values? Assume the domain of g is $[-1, 7.6]$.

Explanation. Since g is continuous on the closed interval $[-1, 7.6]$, g attains its absolute extreme values at either critical points or endpoints. But we just saw that $g'(x) = f(x)$, and so the critical points of g are just the points where $f(x) = 0$. Thus, the critical points of g are approximately 3.1 and 6.25.

Problem 6

Since $f = g'$, g is increasing when f is positive and g is decreasing when f is negative. So we have that $x = 3.1$ is a local maximum while $x = 6.25$ is a local minimum of g . This takes care of the local extreme values.

To find the absolute extreme values of g , we need to determine what are the largest and smallest values among $g(-1)$, $g(3.1)$, $g(6.25)$, and $g(7.6)$. And do not forget the $g(x)$ denotes the (signed) area of the region bounded by the curve and the x -axis.

Clearly $g(-1) = 0$ since a line segment does not have any area. At any time after $x = -1$ we have a positive (net) area. Thus, our absolute minimum occurs at $x = -1$. To find the absolute maximum of g , we need to compare $g(3.1)$ and $g(7.6)$. We see that we have the greatest area under the curve at $x = 3.1$ because by the time we get to $x = 7.6$, the area we have subtracted off between $x = 3.1$ and $x = 6.25$ is greater than what we have added back on from $x = 6.25$ to $x = 7.6$. Therefore, our absolute maximum occurs at $x = 3.1$.

[4]

- (d) Where is the graph of g concave up? Concave down?

Explanation. g is concave up when f is increasing, which is on $(-1, 0) \cup (4.25, 7.6)$. Similarly, g is concave down when f is decreasing, which is on $(0, 4.25)$. Remember, $g' = f$.

[2]